

AgriCare MM

AI plant-disease detection & crop calendar for Myanmar farmers — English + မြန်မာ

Design document: idea flow, system architecture, data model, API, and build roadmap

Prepared August 2026

1. Concept Overview

AgriCare MM is a mobile-first web app for farmers with two core features:

Feature	What it does	Value to the farmer
Plant Doctor (disease detection)	Farmer photographs a sick plant. AI vision identifies the disease and returns the diagnosis, cause, and treatment steps in both English and Burmese .	Instant expert advice without traveling to an agriculture office; understandable in the farmer's own language.
Crop Calendar (planting tracker)	Farmer records what they planted and when (e.g. rice on 15 June). The app computes the expected harvest window, sends reminders, and keeps a year-by-year cultivation history.	Never miss the harvest window; see past seasons at a glance to plan the next one.

Why these two together: the calendar tells the app what the farmer is growing and at which growth stage the crop is. That context makes disease detection more accurate (rice at tillering stage has different likely diseases than rice near harvest) and lets treatment advice reference the farmer's actual timeline.

2. Feature 1 — Plant Doctor (photo → diagnosis) Flow



Figure 1 — Plant Doctor user flow. Green = user steps, blue = AI/backend steps, amber = validation guard.

Bilingual output — example

Field	English	မြန်မာ
Disease	Rice Blast (leaf blast)	စပါးဂုတ်ကျိုးရောဂါ
Cause	Fungus Magnaporthe oryzae; spreads in humid weather and over-fertilized fields.	မှိုကြောင့်ဖြစ်ပြီး စိုထိုင်းဆများသောရာသီဥတုနှင့် ဓာတ်မြေဩဇာအလွန်အကျွံသုံးသောလယ်များတွင် ပျံ့နှံ့သည်။
Treatment	1. Drain the field for 3–4 days. 2. Apply tricyclazole fungicide. 3. Reduce nitrogen fertilizer.	၁။ လယ်ရေကို ၃-၄ ရက် ဖောက်ထုတ်ပါ။ ၂။ မှိုသတ်ဆေး (tricyclazole) ဖြန့်ပါ။ ၃။ နိုက်ထရိုဂျင် မြေဩဇာလျှော့သုံးပါ။

The AI is prompted to answer in structured JSON with parallel `en` and `mm` fields, so the UI can toggle language instantly without a second API call.

3. Feature 2 — Crop Calendar Flow

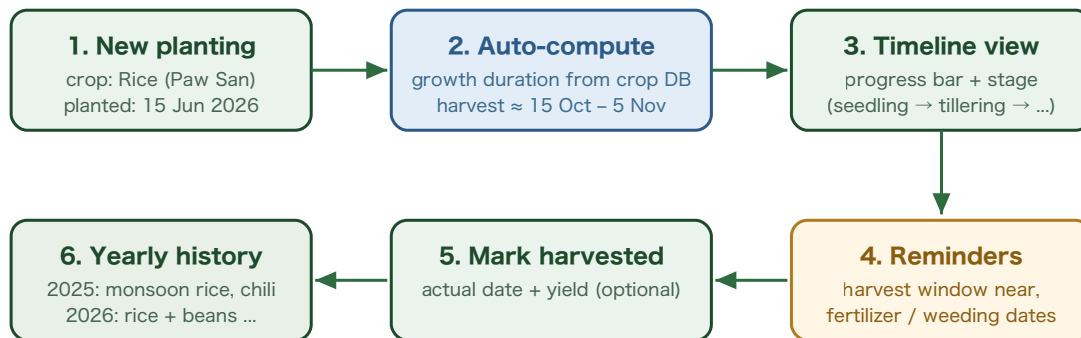


Figure 2 — Crop Calendar flow. The crop database stores typical growth durations (rice ≈ 120–140 days, chili ≈ 90 days, ...) to predict the harvest window.

Yearly history — what the farmer sees

Year	Crop	Planted	Expected harvest	Actual harvest	Notes / diagnoses
2026	Rice (Paw San)	15 Jun 2026	15 Oct – 5 Nov	— (growing, day 75)	1 diagnosis: leaf blast, treated 28 Aug
2026	Chili	2 Feb 2026	~3 May	7 May 2026	no disease
2025	Rice (monsoon)	20 Jun 2025	~25 Oct	30 Oct 2025	brown spot, treated

4. System Architecture

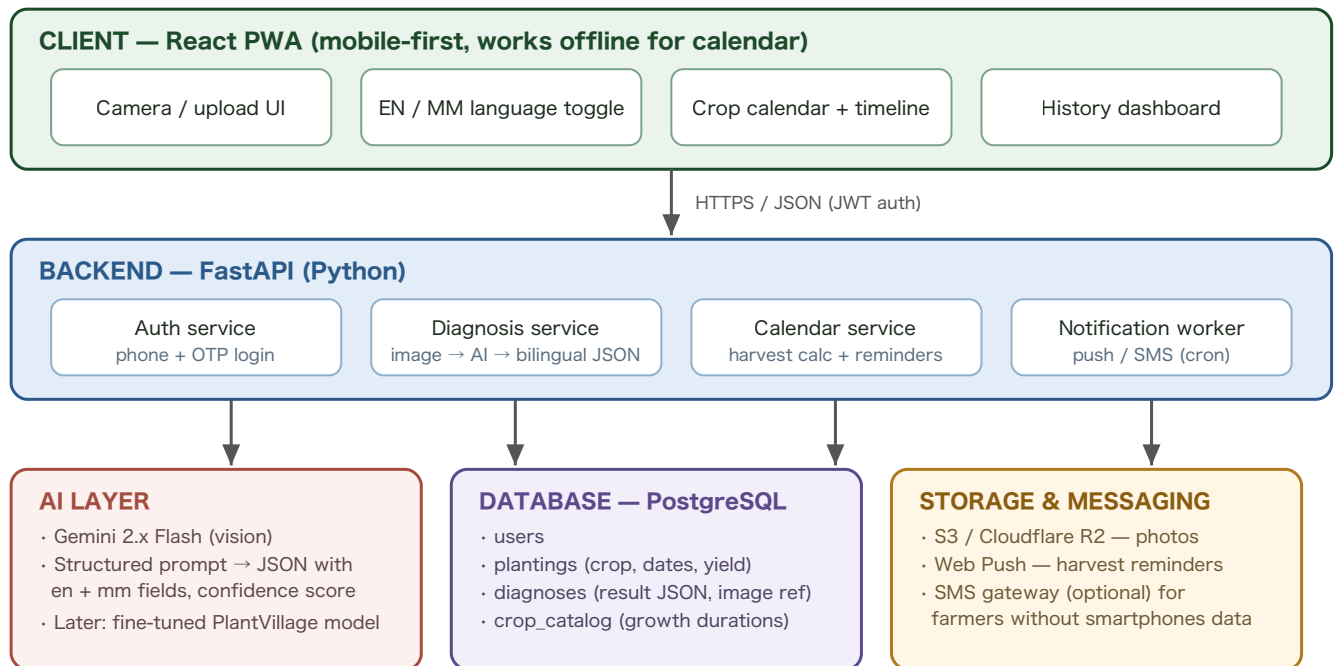


Figure 3 — System architecture. One backend, three supporting layers. Everything deployable on a single VPS or free-tier cloud to start.

5. Data Model

Table	Key columns	Notes
users	id, phone, name, village, preferred_lang (en mm), created_at	Phone + OTP login — most Myanmar farmers have a phone number but not email.
crop_catalog	id, name_en, name_mm, min_days, max_days, stages_json	Seeded reference data: rice 120–140 days, chili ~90, groundnut ~110, etc. Drives harvest prediction and growth-stage display.
plantings	id, user_id, crop_id, variety, planted_on, expected_harvest_start, expected_harvest_end, actual_harvest_on, yield_amount, season_year, notes	One row per field/crop per season. season_year powers the yearly history view.
diagnoses	id, user_id, planting_id (nullable), image_url, disease_en, disease_mm, confidence, result_json, created_at	result_json holds the full bilingual AI answer (cause, treatment steps, prevention). Linking to a planting builds a disease timeline per crop.

6. API Endpoints

Method	Endpoint	Purpose
POST	/auth/otp · /auth/verify	Request OTP, verify and receive JWT.
POST	/diagnoses	Multipart image upload (+ optional planting_id). Returns bilingual diagnosis JSON.
GET	/diagnoses?year=2026	Diagnosis history for the user.
GET	/crops	Crop catalog for the "new planting" form (bilingual names).
POST	/plantings	Create planting; server computes expected harvest window.
PATCH	/plantings/:id/harvest	Record actual harvest date and yield.
GET	/plantings?year=2026	Yearly cultivation history.

AI prompt contract (Diagnosis service): the backend sends the photo plus context ("crop: rice, planted 75 days ago, region: Ayeyarwady") and requires this JSON shape:

```
{ "is_plant": true, "disease": {"en": "...", "mm": "..."}, "cause": {"en": "...", "mm": "..."}, "treatment_steps": [{"en": "...", "mm": "..."}], "prevention": {"en": "...", "mm": "..."}, "severity": "low|medium|high", "confidence": 0.87 }
```

If `is_plant` is false or confidence < 0.6, the app asks the farmer for a clearer photo instead of showing a shaky answer.

7. Tech Stack

Layer	Choice	Why
Frontend	React + Vite, PWA, Tailwind CSS	Installable on the phone home screen; calendar works offline; camera via standard web APIs.
Backend	FastAPI (Python)	Fast to build, async image handling, easy AI SDK integration.
AI vision	Gemini 2.x Flash (or GPT-4o)	Strong multilingual output — generates Burmese natively, so no separate translation service is needed.
Database	PostgreSQL (Supabase free tier)	Relational fit for plantings/diagnoses; Supabase also gives auth + storage if you want fewer moving parts.
Image storage	S3 / Cloudflare R2 / Supabase Storage	Cheap, presigned uploads keep the backend light.
Notifications	Web Push; SMS later	Harvest-window and treatment follow-up reminders.
Fonts / i18n	Noto Sans Myanmar; UI strings in <code>en.json</code> / <code>mm.json</code>	Correct Burmese rendering everywhere; language toggle is instant.

8. Build Roadmap

Phase	Deliverable	Definition of done
1. MVP core (week 1)	Photo upload → bilingual diagnosis result screen (no login; anonymous session).	A rice-leaf photo returns correct disease + treatment in EN and MM.
2. Calendar (week 2)	Crop catalog, new-planting form, harvest prediction, timeline view.	Planting rice on a date shows the correct expected harvest window and growth stage.
3. Accounts & history (week 3)	Phone OTP login, yearly history dashboard, diagnoses linked to plantings.	A returning user sees 2025 vs 2026 seasons side by side.

4. Polish (week 4)	Push reminders, offline calendar (PWA), low-confidence retake flow, Burmese UI review by a native speaker.	Demo-ready: full farmer journey from photo to harvest record.
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Risks to watch:

- **Burmese quality** — LLM Burmese is good but review agricultural terms with a native speaker; keep a glossary of corrected terms and inject it into the prompt.
- **Connectivity** — rural data is slow; compress images client-side (~500 KB) before upload and cache the calendar offline.
- **Wrong advice** — always show confidence and add a disclaimer to consult the township agriculture office for severe cases; never recommend specific pesticide dosages without a verified source.